



Validating an ullage detection technique for liquid propellant tanks

Justin Wheeler

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

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Knowledge of location of the ullage, the gas pocket within a liquid propellant tank, and ullage dynamics is of vital importance to deep-space refueling depots. Characterizing the size and location of the ullage within liquid propellant tanks in equilibrium is crucial to the Mars mission. An experiment payload was designed to test a method of gauging the size and location of the ullage within a liquid propellant tank. Broadband white noise is passed into a stainless steel tank filled with a rocket propellant simulant. The acoustic vibrations are measured at several point sensor locations along the surface where changes in amplitude are representative of changes in the thickness of the liquid layer coating the inside of the tank. The data suggests that the sensors are able to detect the passage of the liquid surface under the effect of gravity but are not accurate enough for space flight at the moment.

I. INTRODUCTION

Knowledge of the location of the gas within liquid propellant tanks in space is of vital importance. Gas presence at the exit of the tank when fluid is transferred can cause harmful spikes in pressure damaging sensitive equipment. Most spaceflight endeavors involve a liquid propellant tank being drained to completion before the liquid can settle into low gravity equilibrium. With the efforts to put humans on Mars, refueling depots in space are necessary. The liquid propellant within these refueling tanks will have sufficient time to settle into their lowest energy state.

In the absence of gravity, capillary force keeps the liquid stuck to the inner surface of the propellant tank leaving a gas bubble floating in the center. The dynamics of this ullage have not yet been characterized in low gravity.

An experiment, named Microgravity Ullage Detection (MUD), was designed by the primary author and built by students at Carthage College. The payload is able to operate and collect data on Earth with the effects of gravity and can operate aboard the ZeroG Inc. modified Boeing 727 aircraft as well as the Blue Origin New Shepard payload lockers in the absence of gravitational effects. Data collected and analyzed in this experiment is obtained on Earth and will be used to validate that this method works as intended before collecting data in space.

II. THEORY

Acoustically driven vibrations propagate through surfaces as waves. On cylindrical tanks, these waves propagate axially, radially, and circumferentially [1]. The walls of the tank can be considered an infinitely coupled series of simple harmonic oscillators with frequencies ranging

across all available frequencies. According to the Debye model, each atom in a solid is treated as a simple harmonic oscillator with a certain vibrational frequency. In the Debye model, the vibrational frequencies of these oscillators are distributed across a range, forming a continuous spectrum. This is in contrast to the simpler Einstein model, which assumes that atoms in a material do not vibrate independently of one another [2].

Some frequencies will be faintly present in a material, whereas others will be at large amplitudes. This is because, an external force applied at the natural frequency of a harmonic oscillator within the material induces resonance. This leads to oscillations with large amplitudes because the energy input is synchronized with the natural frequency of the oscillators. An oscillating atom has a natural frequency

$$\omega_0 = \sqrt{\frac{k}{m}}, \quad (1)$$

where ω_0 is the natural frequency, k is the spring constant related to the stiffness of the material, and m is the mass of the atom [3]. Therefore, the amplitudes of these atoms acting as harmonic oscillators depend on the material's stiffness and mass. On top of this, the geometry of the material is important, as the Debye model indicates, the quantum oscillators are interacting with one another, and defects in crystalline structure play a large role in influencing the resonant frequencies of a material.

As the mass and effective stiffness of the tank decrease, the frequency and amplitude of the overall oscillations increase. If a cylindrical propellant tank is filled with liquid, the vibrations are damped at locations that are covered by liquid within the tank [4]. This is because the liquid presence can be approximated as additional mass added to the tank wall, shifting the resonant frequencies and amplitudes. The amount of damping is proportional to the thickness of the liquid layer at each location.

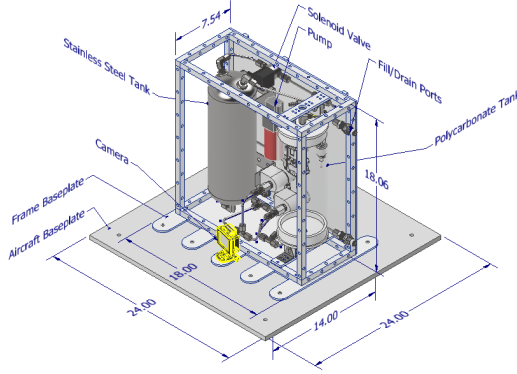


FIG. 1. Computer Aided Design model of the Microgravity Ullage Detection payload with relevant key mechanical features identified.

In the presence of gravity, this is a binary system, either the patch is covered or it is not covered. The thickness of the liquid does not change at any point on the tank, liquid is effectively only present or not. Proof of concept on Earth is necessary before testing in space. In the absence of gravity, capillary forces keep the liquid covering the inner surface of the tank with the gas forming a bubble floating in the center, known as the ullage. The distance between the ullage and the tank wall is all liquid, the thickness of which is proportional to the damping force at that location. Thus, by driving the tank with a wide enough range of frequencies to excite the modes of the tank, and sensing the excitation in waves propagating along the outside surface, one can deduce information about the thickness of liquid covering the inside surface of the tank at the locations of the sensors. If the entire tank is covered with sensors, the size and location of the ullage can be reconstructed from the data taken by the sensors.

The theory that will be investigated in this paper is that under the effect of gravity, the liquid-vapor interface separating the ullage from the liquid can be detected by observing the change in amplitude of the vibrations at locations along a stainless steel tank during fluid transfer.

III. METHODS

A. Experiment Payload Design

The experiment payload is designed to measure the size and location of the ullage within a stainless steel tank during fluid transfer with several highly sensitive piezoelectric patches acting as strain gauges. The hardware is controlled through a Raspberry Pi 4 microcontroller with a data acquisition device attached.

The fluid used in this experiment is a hydrofluoroether (HFE) named Novec 7100. This fluid mimics the sur-

face tension of Monomethylhydrazine, a common liquid propellant used in rockets. The surface tension is an important characteristic of the liquid propellant to match because it is the property that most greatly affects the shape of the ullage in microgravity. Additionally, HFE is nontoxic and compatible with the stainless steel tank making it a safe substance for lab use.

The mechanical design of the payload includes a flow loop consisting of a pump and several solenoid valves that can transport fluid into and out of a stainless steel tank. The tank operates as a scaled-down liquid propellant tank aboard a refueling depot in space. Pumping out of the stainless steel tank simulates a refueling depot transferring fluid into a third-party spacecraft. A polycarbonate “catch tank” collects the fluid as it drains from the stainless steel tank. This tank serves purely as a means of visualization for the fluid remaining in the stainless steel tank and storage of the transferred propellant. All vibration data collected comes from the stainless steel tank as it rings well with acoustic noise.

Broadband white noise across 500 to 5 kHz is passed into the stainless steel tank and the vibrations are picked up by sensor patches. The sensor patches are adhered to the stainless steel tank with double-sided tape to allow mechanical strain in the tank wall to be transferred as directly as possible to the sensors.

The sensors used on this experiment are made of a macro fiber composite material, Lead Zirconate Titanate (PZT). Macro fiber composites are made up of piezoelectric fibers embedded in a polymer matrix. When subjected to mechanical stress, the piezoelectric fibers within the PZT patch generate an electrical charge proportional to the applied force. This charge can be measured and represents the amplitude of vibration at a given location along the applied surface [5]. Additionally, when an electric potential is placed across terminals of the piezoelectric patch, the patch mechanically deforms [6]. These patches can act as either actuators which induce vibration or sensors which measure vibration depending on how they are configured.

The PZT patches are configured as shown in fig. 2. There are eight patches on the stainless steel tank. The topmost patch acts as an actuator and injects the broadband white noise into the tank. The seven patches beneath the actuator are wired to the data acquisition device and are used for ullage detection.

The data acquisition device is a MCC 118 DAQ HAT for Raspberry Pi [7]. This is a voltage measurement device attached to the top of the Raspberry Pi microcontroller. This external hardware is required to perform analog measurements of the PZT patches as the Raspberry Pi does not have a built-in analog-to-digital converter. The MCC 118 DAQ HAT has 8 channels with a 100 kHz sample rate spread evenly across the channels which yields an 11.1 kHz sample rate per channel if all channels are being used. This is equivalent to an uncertainty in time measurements of

$$\delta_t = 9 \times 10^{-5} \text{ s.} \quad (2)$$

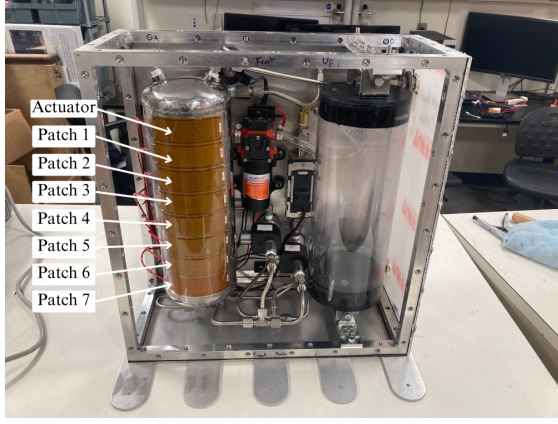


FIG. 2. MUD Payload with PZT Patches labeled. The patch numbers increase as they go down the tank.

Additionally, the data acquisition device has 12-bit resolution which means the finest divisions in voltage that can be measured is 2 mV yielding an uncertainty in voltage measurements of

$$\delta_V = 2 \text{ mV}. \quad (3)$$

An accurate knowledge of the location of the ullage is required to verify that PZT patches can predict its location. Measurements of the volume of liquid in the tank and the flow rate of HFE allow us to calculate where the true location of the ullage is. During fluid transfer, the fluid is transferred through a Keyence FD-H10 flow sensor. The flow sensor operates under the Doppler effect, leveraging ultrasonic waves to measure the flow velocity of liquids. A transmitter in the FD-H10 emits ultrasonic waves directed towards the flowing liquid, and these waves are reflected by suspended gas bubbles within the liquid. The movement of these particles shifts the frequency of the wave that gets reflected by a reflective surface, which is detected by a receiver. The magnitude of the frequency shift is proportional to the flow velocity [8]. This flow sensor was selected because it operates well with two-phase flow. As the ullage location is unknown, if there is gas present near the sump of the tank, there will be bubbly gas-liquid flow through the flow sensor.

B. Calculations

The flow sensor has an analog output that outputs at a range of electrical currents proportional to the flow rate. The analog lower output limit I_{low} corresponds to 0 L/min and the analog upper limit I_{high} corresponds to 3 L/min. A simple linear mapping relationship between the current output measured by the data acquisition device to flow rate Q measured by the flow sensor for any outputted current I at any given time t is given by

$$Q(I(t)) = \frac{3(I(t) - I_{\text{low}})}{I_{\text{high}} - I_{\text{low}}}. \quad (4)$$

This relationship comes from a linear bijection between two closed intervals [9]. For the applications of this experiment, the range of analog output is 4 – 20 mA.

The data acquisition device, however, measures voltage and not current. Therefore, it is necessary to pass the output current through a resistor leading to ground. The voltage drop across the resistor is measured by the data acquisition device to convert the flow sensor output current to a measurable output voltage. Recalling Ohm's law the current flowing through a resistor with resistance R

$$I = \frac{V}{R} \quad (5)$$

where V is the voltage drop across the resistor. Combining eq. (4) and eq. (5) yields the following relation between flow rate and measured voltage

$$Q(V(t)) = \frac{3\left(\frac{V(t)}{R} - I_{\text{low}}\right)}{I_{\text{high}} - I_{\text{low}}}. \quad (6)$$

The total accumulated flow Γ_{out} out of the tank for a duration T is the sum of all the instantaneous flow rates $Q(V(t))$ multiplied by the sampling frequency dt :

$$\Gamma_{\text{out}}(T) = \sum_{t=0}^T Q(V(t))dt. \quad (7)$$

The total volume of fluid remaining in the tank Γ_{rem} after any duration of time is given by

$$\Gamma_{\text{rem}}(T) = \Gamma_0 - \Gamma_{\text{out}}(T), \quad (8)$$

where Γ_0 is the initial volume in the tank.

To calculate the position of the Ullage at any given time, information about the geometry of the tank is necessary. The stainless steel tank shown in fig. 3 which is used in the MUD payload is pill-shaped, with a cylindrical body and two domes capping off the ends. The patches are located on the cylindrical section where the ullage location as a function of time with a constant flow rate is a linear calculation.

For a given tank radius r , and height of the lower end cap H_{cap} the Ullage is located U meters above the bottom of the tank. Recalling the equation for the height of a cylinder with respect to a given radius and volume

$$U(t) = H_{\text{cap}} + \frac{\Gamma_{\text{rem}}(t)}{\pi r^2 c}. \quad (9)$$

where c is the conversion factor for cubic meters to liters. U is representative of the true height of the ullage with respect to the bottom of the inside surface of the tank.

The root mean square (RMS) is a useful tool in analyzing the amplitude of the oscillations in the tank wall as measured by the PZT patches. The root mean square of a set of voltage samples $V_1, V_2, V_3, \dots, V_N$ is given by

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N V_i^2}. \quad (10)$$

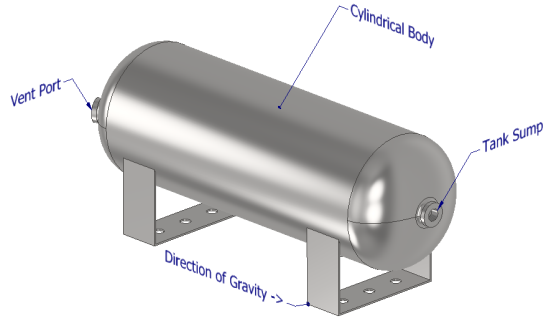


FIG. 3. Computer aided design model of the Stainless steel tank. The geometry of which is used to measure the volume within the tank and the height of the ullage as a function of flow rate.

In order to effectively investigate the amplitude of the voltage signals over time, we take the moving root mean square of the signal. The moving root mean square takes chunks of measured data in discrete window sizes and computes the root mean square of that subset. The window shifts over the data set at discrete intervals to cover the entire range. This method allows us to investigate changes in amplitude without the frequency of oscillation being taken into account. This is desirable because the key difference between a sensor covered by liquid and an uncovered sensor is the amplitude of oscillations.

C. Concept of Operations

We filled the stainless steel tank with 3.00 ± 0.01 L of HFE. The pump was set to flow at approximately 0.75 Liters per minute and the flow sensor was set to record flow rate data. We then turned the actuator on and started the data collection process on the data acquisition device. Shortly after, the pump was turned on and all of the liquid was transferred to the catch tank.

IV. RESULTS

The raw signal output as well as the RMS of the signal from patch 4 during the tank drain is shown in fig. 4. There is a clear amplitude shift around the 120 second mark, with mostly constant signal before and after this shift. All seven patches exhibit this behavior at locations related to where they are positioned on the tank. Patch 4 is relatively close to the center so the amplitude shift occurs near the middle of the drain.

Taking a closer look at the root mean square of the signal from patch 4 in fig. 5 shows the amplitude shift at a greater scale. The Gaussian filter smooths out the high frequency noise in the raw RMS so that we can investigate the overall changes in signal. There is approximately

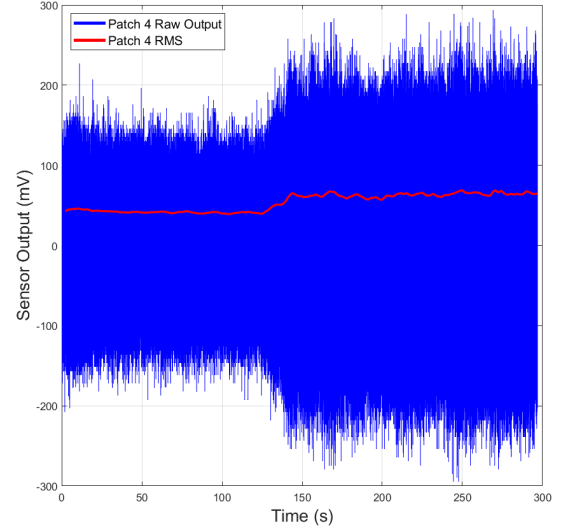


FIG. 4. Plot of the raw signal output and RMS of patch 4 during the drain versus time. The raw signal is oscillating at frequencies driven by the actuator.

a 25 mV increase in amplitude over a period of about 20 seconds. All seven sensor patches exhibit this behavior as well.

The derivative of the smoothed RMS signals from every patch during the drain test is plotted in fig. 6. The point at which the rate of change is greatest is highlighted as well. These points indicate the predicted location of the ullage from our PZT patches. If the theory is true, the

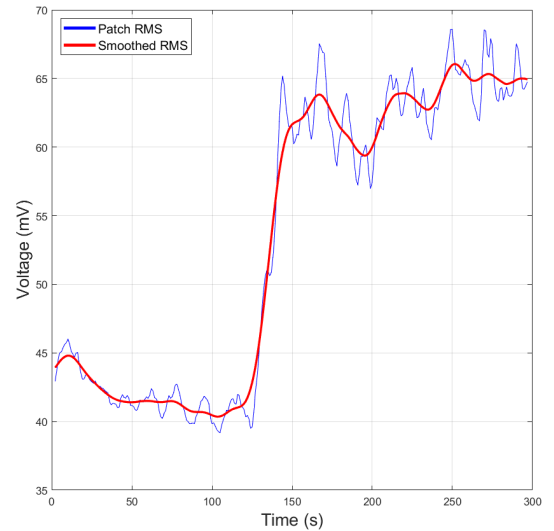


FIG. 5. Root Mean Square (RMS) of the data taken by patch 4 during the drain along with the smoothed curve by a Gaussian filter.

time associated with the highlighted points corresponds to the time at which the ullage boundary is directly in line with the geometric center of the respective PZT patch.

In order to determine how accurate our PZT patches are at predicting the location of the ullage, a precise definition of the ullage location with respect to time is required. The known locations of the patches can then be used to determine the time at which the ullage passed the center of each sensor. The height of the ullage is plotted as a function of time in fig. 7. These values were recorded and will be used to quantify the effectiveness of the PZT patches.

A plot of the true time the ullage passes the sensor versus the predicted time the ullage passes the sensor by the peak in the derivative of the root mean square is shown in fig. 8. The slope m of the calculated line is

$$m = 1.024 \pm 0.005, \quad (11)$$

and in the limit of the ullage being at the center of the patch at $t = 0$ the ullage is predicted to occur -25.4 ± 0.9 seconds before this by the linear regression line. The correlation coefficient of the trendline is

$$R = 0.982. \quad (12)$$

V. DISCUSSION

The large amplitude shift in fig. 4 is important as it is a visual interpretation of the wet and dry sensor patch. It is characteristic of some change in the acoustic structure

of the tank wall, the only thing that has changed during the drain being the lack of liquid covering the sensor patch. Another important aspect of this graph is how relatively constant the amplitudes of the pre-increase and post-increase oscillations are. As the liquid in the tank starts to drain out, there is really no difference in sensor output because the amount of liquid in the small region covering the inner surface of the tank at the patch location is effectively unchanged. However, there is a large shift as the liquid surface moves over the patch and eventually leaves the patch area entirely.

The smoothed root mean square in fig. 5 is important because it shows how there is one and only one large increase in the patch amplitude. An explanation for this increase is the change in the liquid covering the surface where the patch lies inside the tank. This is also visually observable in fig. 6 where all of the rates of change of the patch RMS signals are plotted. In each patch, there is one large peak corresponding to a large increase in amplitude. Additionally, each patch has a large shift in amplitude sequentially in the order which was to be expected. It is important to note however that patch 3 had a relatively small increase followed by another similar size peak. This could be characteristic of the patch being located on a vibrational node that does not shift during the change in mass within the tank. Or, there could be issues with the pump slowing down momentarily.

The height of the ullage as a function of time in fig. 7 is a relatively linear function with a negative slope. The slope begins to decrease in magnitude near the end of the plot which can be explained by less gravitational force

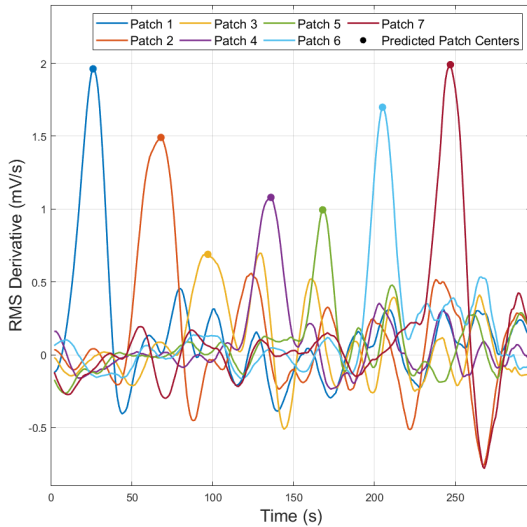


FIG. 6. Derivative of the smoothed RMS outputs of all seven sensor patches during the drain. The peaks are identified as these points predict the time at which the ullage passes each patch.

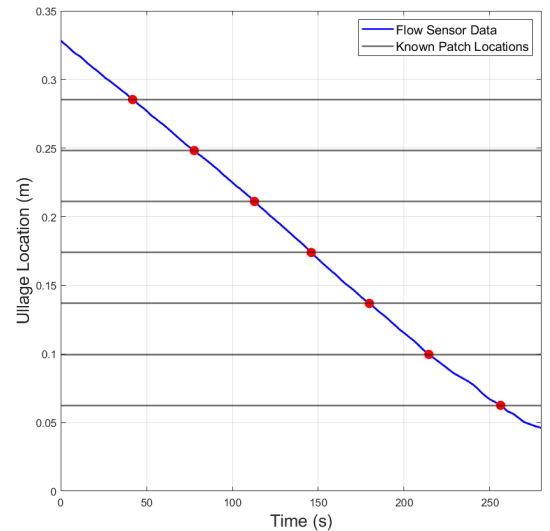


FIG. 7. The distance between the bottom of the tank and the liquid surface as a function of time. The horizontal lines are at the measured patch locations on the tank. The highlighted points at the intersection of the lines are at the true times at which the ullage passed the center of the patches.

acting on the pump due to the decreasing liquid mass in the tank. The true time at which the ullage passes the patch centers is determined by the intersection of the lines.

The final plot [fig. 8] demonstrates the accuracy of the ullage detection technique. A perfectly linear line passing through the origin with a slope of 1 would mean that the PZT patches can perfectly predict the location of the ullage. This measurement likely has a higher uncertainty than assumed in this analysis as the linear regression line does not pass through the origin. However, since the line has a high correlation to the data, this fact can be explained by drift in the flow sensor data.

The flow sensor has varying degrees of accuracy depending on the flow rate. The flow rate used in this experiment (0.75 Liters per minute) was determined to be the most accurate through a series of manual time measurements compared to the flow rate. This is likely the source of the drift in the recorded true values.

VI. CONCLUSION

The data suggests that the point at which the root mean square output of the PZT patches is increasing the greatest is related to the point at which the liquid layer is passing the center of the PZT patch. The method provided a high degree of relative accuracy while yielding a low level of absolute accuracy. The predicted ullage measurements are spaced close to exactly as predicted with relation to the other measurements as indicated by the high correlation coefficient and slope that is close to

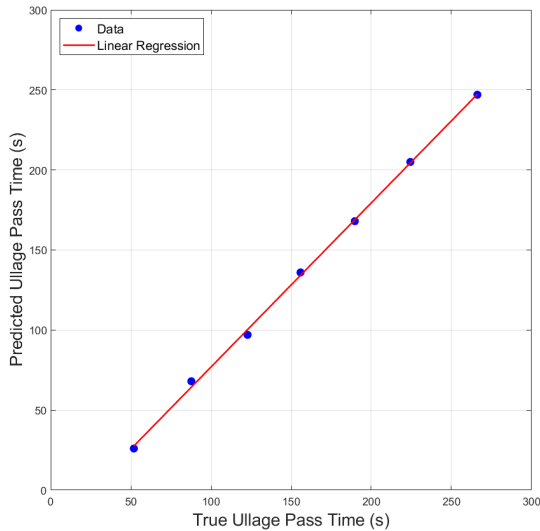


FIG. 8. Plot of the predicted location of the ullage by the RMS peaks versus the known ullage location. A linear least squares regression line will allow us to assess the accuracy of this method.

1. However, they are not completely correct with respect to the true location of the ullage as the linear regression line is far from the origin in the $t = 0$ case.

This research yields promising results for the future of ullage detection in microgravity. Improvements to the accuracy of the flow sensor and known patch locations will yield better results. This experiment is still in progress and will one day be sent on a suborbital mission to gauge its accuracy in the absence of gravity.

Appendix A: Uncertainty Analysis

The uncertainties in individual measurements of length, voltage, and time are as follows:

$$\delta_L = 0.001 \text{ m}, \quad (\text{A1})$$

$$\delta_V = 2 \text{ mV}, \quad (\text{A2})$$

$$\delta_t = 9 \times 10^{-5} \text{ s}. \quad (\text{A3})$$

The uncertainties in the time at which the ullage passes the centers of the sensor patches as predicted by the RMS derivative is

$$\delta_{t\text{-pred}} = 1 \text{ s}. \quad (\text{A4})$$

This was determined through the uncertainties in voltage around the RMS derivative. The peaks are sharp enough and the uncertainties are low enough that a small uncertainty of 1 second was reasonable.

The uncertainties in the true location of the ullage as determined by the flow sensor was also 1 second. This was determined with similar reasoning as above.

The slope m and y-intercept b of the linear least squares regression line are as follows:

$$m = \frac{\langle (x - \bar{x})(y - \bar{y}) \rangle}{\langle (x - \bar{x})^2 \rangle}, \quad (\text{A5})$$

$$b = \bar{y} - m\bar{x}. \quad (\text{A6})$$

These are determined by attempting to find a function that minimizes the squared difference in y-distance between the data points and the functions.

The uncertainties in the slope and y-intercept are determined as follows:

$$\delta_m = \frac{\delta_y}{\sigma_x \sqrt{N}}, \quad (\text{A7})$$

$$\delta_b = \frac{\sigma_y}{\sqrt{N}} \sqrt{1 + \left(\frac{\bar{x}}{\sigma_x} \right)^2}. \quad (\text{A8})$$

Where σ_x is the statistical standard deviation of the population of x and δ_x is the uncertainty in each measurement of x . It is important to note that these formulas are assuming that the uncertainties in each quantity are constant for each measurement.

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